

Xianhao Carton Liu

📧 Carton22
✉ liu03008@umn.edu
🌐 carton22.github.io

Education

Computer Science Ph.D. Student

Advisor: **Zhu-Tian Chen**

Research topic: Human AI Interaction in Physical Environments via Visualizations, XR and Robotics

University of Minnesota, Twin Cities

Aug 2024 – Now

B.Eng. with Honors in Artificial Intelligence (Turing Class)

Outstanding honored undergraduate in **Chu Kochen Honors College**

Zhejiang University

Sep 2020 – June 2024

Research Experience

Harvard Visual Computing Group, Harvard University

Research intern advised by **Zhu-Tian Chen** and **Hanspeter Pfister**

Cambridge, USA

Jan 2024 - May 2024

Stanford HCI Group, Stanford University

Research intern advised by **Jackie Yang** and **James Landay**

California, USA

Jul 2023 - Sep 2023

ZJU International Design Institute, Zhejiang University

Research intern advised by **Lingyun Sun** and **Zejian Li**

Hangzhou, China

Mar 2022 - Jun 2023

Publications

Can AR Embedded Visualizations Foster Appropriate Reliance on AI in Spatial Decision Making? A Comparative Study of AR X-Ray vs. 2D Minimap

Accepted in CHI 2026 [Paper] [Video] [Github]

Jan 2026

• [Xianhao Carton Liu](#), [Difan Jia](#), [Tongyu Nie](#), [Evan Suma Rosenberg](#), [Victoria Interrante](#), [Chen Zhu-Tian](#)

Reality Proxy: Fluid Interactions with Real-World Objects in MR via Abstract Representations

Accepted in UIST 2025 [Paper] [Video]

May 2025

• [Xiaoan Liu](#), [Difan Jia](#), [Xianhao Carton Liu](#), [Mar Gonzalez-Franco](#), [Chen Zhu-Tian](#)

Vitesse Link: A Mixed Reality and 2D Display Hybrid Approach for Visual Analysis of 3D Tissue Maps

Submitted to Nature Methods [Github]

Sep 2025

• [Eric Mörrth](#), [Morgan Turner](#), [Cydney Nielsen](#), [Xianhao Carton Liu](#), ... , [Chen Zhu-Tian](#), [Nils Gehlenborg](#)

Projects

WebXR Fundamental 3D Interaction Methods for Object Manipulation

[Video] [Github]

• A WebXR interaction framework for object manipulation methods including translation, rotation and resizing.

WebXR for 3D interaction

Spring 2024

Generative XR Spatial Interfaces for Human Robot Co-exploration

[Video] [Github]

• Reconstructing and synthesizing real-world scene via mobile robot on real-time for human space interaction

Situated XR Interfaces

Summer 2025

Human-AI decision-making in spatial environment via AR visualizations

[Video] [Github]

• Understanding human overreliance and trust on AI in spatial decision-making via different AR visualizations

Human AI Collaboration

Spring 2025

Handbot——Apple WWDC2023 Swift Student Challenge

[Video] [Github]

• A swift playground encouraging critical thinking and editing on LLM and other genAI content in multi-modalities.

Human-centered GenAI

Apr 2023

Is raining threes worth it?——NBA 3-pointer Data Visualization

[Video] [Website]

• An interactive visualization website featuring NBA 3-pointers data and key players, received 17000+ views

Data Visualization

Dec 2022

Skills

Programming

WebXR/Unity3D/Swift/ROS2/ROS1/Pytorch/ThreeJS/React/C#/C/C++/Python

HCI Methods

Quantitative Analysis/Interview Thematic Analysis/Participatory Design and Analysis

Tools

Apple Vision Pro/Meta Quest/Turtlebot4/Nvidia Jetson/RasPi4/Figma/Xcode/Pr/PS